

CRATERS

The vast majority of lunar craters are produced by impacts. Asteroids usually strike the Moon at velocities of about 45,000 mph (72,000 kph). The resulting crater is about 15 times larger than the impacting body. Unless the asteroid nearly skims the surface on entry, the resultant crater is circular. Three types are formed. Those smaller than 6 miles (10 km) across are bowl-shaped, having a depth of around 20 percent of the diameter. Craters between 6 and 90 miles (10–150 km) in diameter have outer walls that have slumped into the initial crater pit. There is often a central mountainous peak produced by the recoil of the underlying stressed rocks. The crater depth is a few miles, and a lot of excavated material falls back into the crater just after the impact. Craters wider than 90 miles (150 km) contain concentric rings of mountains, created as rebounding material rippled out from the center before solidifying. Such craters were so deep that hot magma flooded to the surface and filled the bottom of the crater with lava.

RAY CRATERS

Material ejected from a crater during an impact is often confined to narrow jets. Where this material hits the surface, it plows up the lunar soil, and this disturbed region then reflects more sunlight than its surroundings. From Earth, these appear as rays. The rays around Tycho Crater (far right) extend for thousands of miles.



SUNRISE OVER COPERNICUS CRATER



1 Just after dawn in the crater, the low eastern Sun casts long shadows, which emphasize the variation in height between the floor and rim.



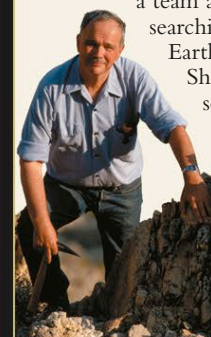
2 Halfway through the morning, small shadows enhance the ejecta blanket outside the crater. The temperature inside the crater is rising.



3 At noon, the Sun is overhead, and the scene appears much flatter and washed out. The temperature is now more than 212°F (100°C).

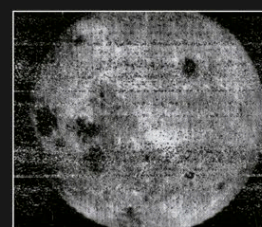
EUGENE SHOEMAKER

Gene Shoemaker (1928–1997) was an American astrogeologist who studied terrestrial and lunar meteorite impact craters and dreamed of going to the Moon. Addison's disease prevented that. Instead, he taught the Apollo astronauts to be field geologists. In 1969, he joined a team at Palomar, searching for near-Earth asteroids. After Shoemaker died, some of his ashes were carried to the Moon aboard the Lunar Prospector space probe in 1999.

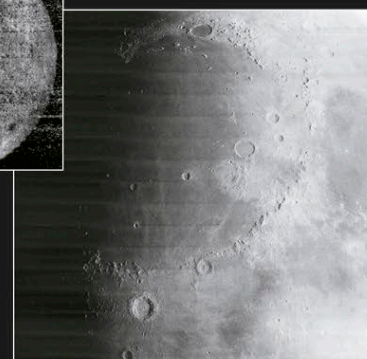


MAPPING THE MOON

Some ancient Greeks thought that the Moon was like the Earth and that its dark areas were water. This belief continued into the 17th century, when the dark patches were given aquatic names such as mare (sea) and oceanus (ocean) on the first proper maps. Palus Putredinis (the Marsh of Decay) and Sinus Iridum (the Bay of Rainbows) are evocative examples. Italian astronomer Galileo Galilei was the first to realize that the height of surface features could be added to maps by noting how the shadow lengths changed during the lunar day. The first photographic atlas appeared in 1897, but the real leap forward came with the advent of spaceflight. In 1959, the Soviet Union sent the Luna 3 space probe behind the Moon to photograph the far side. NASA's five Lunar Orbiter spacecraft imaged 99 percent of the lunar surface in 1966–1967, paying special attention to potential Apollo landing sites. In the 1990s, the Moon's mineral composition was surveyed by Clementine and Lunar Prospector. Since 2009, Lunar Reconnaissance Orbiter (LRO) has been engaged in a detailed mapping project.



LUNA 3
On October 7, 1959, the Soviet Union's Luna 3 space probe imaged the far side of the Moon. It had never been seen before.



LUNAR ORBITER IV
This superb wide-angle image of the half-lit Mare Imbrium was one of 546 images taken by NASA's orbiter on May 11–26, 1967, from a height of about 2,485 miles (4,000 km).

GALILEO SKETCHES

Galileo's first telescopic observations of the Moon were made on November 30, 1609. The pictures, published in *Sidereus Nuncius* in 1610, emphasized the roughness of the surface.



SMART-1

During its approach phase, ESA's SMART-1 spacecraft took this image of an illuminated region of the far side, near the lunar north pole, on November 12, 2004, from a distance of about 37,250 miles (60,000 km).